WHY ARE MY ORDERS GETTING DELAYED?

Area: Weather Analysis

Observation: Snowy weather recorded the highest average delivery time followed by rainy, foggy, windy and clear conditions.

Insights: Adverse weather conditions significantly increase delivery delays, indicating weather as a major operational risk factor affecting delivery efficiency.

Recommendation: Introduce weather adjusted ETA'S (Estimated Time of Arrival) and optimise delivery rider allocation during adverse weather conditions.

Area: Traffic Analysis

Observation: High traffic results in increased average delivery time followed by medium and low traffic.

Insights: Traffic congestion emerged as one of the strongest contributors to delivery delays and operational inefficiencies.

Recommendation: Implement predictive routing and increase rider allocation during high-traffic periods.

Area: Time Of The Day Analysis

Observation: Evening deliveries showed the highest average delivery time.

Insights: Evening hours represent peak operational congestion due to increased order volume and traffic intensity.

Recommendation: Increase delivery workforce during evening peak hours.

Area: Courier Experience

Observation: Couriers with 5–9 years of experience recorded lower average delivery times compared to less experienced couriers.

Insights: Experienced couriers demonstrate better delivery efficiency due to route familiarity and operational expertise.

Recommendation: Prioritize experienced riders during high-demand and high-traffic delivery windows.

Area: Preparation Time vs Time of Day

Observation: Average preparation time was highest during evening hours.

Insights: Restaurants experience operational overload during evening peak hours, leading to slower order preparation.

Recommendation: Improve kitchen staffing and streamline restaurant workflows during peak demand periods.

Area: Vehicle Performance

Observation: Car deliveries recorded the highest average delivery times overall.

Insights: Larger vehicles are more impacted by urban traffic congestion and mobility limitations.

Recommendation: Prefer bikes and scooters in dense traffic regions for faster delivery execution.

Area: Vehicle Performance by Time

Observation: Cars experienced especially high delivery times during evening hours.

Insights: Peak-hour traffic congestion affects car-based delivery operations.

Recommendation: Reduce dependency on cars during evening delivery windows and optimize vehicle allocation strategies.

Area: Night-Time Delivery Anomaly

Observation: Bikes recorded unexpectedly higher average delivery times during nighttime hours.

Insights: Nighttime bike delivery inefficiencies may indicate operational or environmental challenges affecting two-wheeler performance.

Recommendation: Investigate nighttime delivery conditions and improve safety/logistics support for bike riders.

KEY INSIGHTS

- High-traffic conditions **increased average delivery times by nearly 18%** compared to low-traffic conditions, establishing traffic congestion as a primary operational bottleneck.

- Snowy weather **conditions increased average delivery times by approximately 26%** compared to clear weather, highlighting adverse weather as a significant operational risk factor affecting delivery efficiency.

- During evening peak hours, experienced couriers (5–9 years) completed deliveries **approximately 16% faster** than less experienced riders, indicating operational expertise significantly improves delivery efficiency under high-demand conditions.